Network Design Paper

MediFleet Logistics 7793 Schimmel Terrace Little Rock, AR 72103 May 5, 2025

To: MediFleet Logistics

Operations Manager

7793 Schimmel Terrace

Little Rock, AR

We are pleased to submit this comprehensive network proposal for MediFleet Logistics, a medical shipping company that requires a secure, reliable, and scalable networking solution. Our design integrates physical, logical, and security layers to support the mission-critical operations of medical product transportation. Our proposed solution will ensure the efficiency and integrity of shipping logistics, from warehouse to customer delivery, while maintaining strict compliance with HIPAA and other regulatory standards.

Sincerely,

Jared Pierce

Network Design Proposal for MediFleet Logistics

1. Executive Summary MediFleet Logistics requires a robust network to support its logistics operations across a central office, multiple warehouses, and fleet vehicles equipped with tracking technology. Our network solution includes wired and wireless infrastructure, cloud integration, and enterprise-grade cybersecurity. This proposal outlines the design, specific hardware, software, and estimated costs to implement a complete network solution.

2. Physical Network Design

- **Headquarters (Little Rock, AR)**
    - Data Center Room
    - Wired connections (Cat6) to each workstation and server
    - Wireless Access Points (WAPs) are available throughout for mobile devices
    - Rack-mounted networking equipment
- **Warehouses (3 locations)**
    - Structured cabling with network cabinets
    - POE (Power over Ethernet) switches for cameras and access points
    - WAPs for scanners and handheld devices
- **Fleet Vehicles**
    - Cellular routers with GPS and an encrypted VPN tunnel back to HQ

3. Logical Network Design

- **VLAN Segmentation**
    - VLAN 10: Admin
    - VLAN 20: Logistics/Operations
    - VLAN 30: Guest Wi-Fi
    - VLAN 40: Security/Surveillance
- **IP Schema**
    - 192.168.10.0/24 – Admin
    - 192.168.20.0/24 – Logistics
    - 192.168.30.0/24 – Guest
    - 192.168.40.0/24 – Surveillance
- **Routing & Switching**
    - Core Layer 3 Switch for internal routing
    - Static routing for simplicity and security

4. Security Aspects

- **Firewall**
    - Fortinet FortiGate 100F with UTM services
- **Intrusion Detection and Prevention**

  Snort IDS/IPS for internal traffic analysis

  Endpoint Security
    - CrowdStrike Falcon for endpoint detection and response (EDR)
- **Email and Web Filtering**
    - Barracuda Email Security Gateway

- **VPN Access**
    - FortiClient SSL VPN for remote users
- **Multi-Factor Authentication (MFA)**
    - Duo Security integrated with Active Directory

5. Device List and Estimated Costs

Device/Software	Quantity	Unit Price	Total Cost
Cisco Catalyst Switch	5	$3,000	$15,000
Fortinet FortiGate 100F Firewall	1	$2,800	$2,800
Ubiquiti UniFi Access Points	12	$200	$2,400
Dell PowerEdge R750 Server	2	$6,000	$12,000
Cellular Routers for Fleet Vehicles	20	$600	$12,000
Barracuda Email Security Gateway	1	$3,000	$3,000
CrowdStrike Falcon (Annual License)	50	$80	$4,000
Duo Security MFA Licenses	50	$3/user	$150
Structured Cabling Materials	-	-	$7,000
Rack and Power Distribution Units	3	$1,000	$3,000
Miscellaneous (patch panels, trays)	-	-	$1,500

Total	**$60,850**

6. Installation and Labor Cost Estimate

- Network Design and Planning: 60 hours @ $150/hr = $9,000
- Equipment Installation: 120 hours @ $125/hr = $15,000
- Software Configuration and Testing: 50 hours @ $140/hr = $7,000
- Staff Training and Documentation: 20 hours @ $100/hr = $2,000
- **Total Installation Cost: $33,000**

7. Timeline and Deployment Strategy

- Week 1–2: Procurement and cabling installation
- Week 3–4: Device installation and configuration
- Week 5: Security testing and compliance check
- Week 6: Staff training and system hand-off

8. Conclusion This network solution provides MediFleet Logistics with a secure, reliable, and scalable IT infrastructure ready to support daily operations and future growth. Our approach ensures minimal downtime, regulatory compliance, and full operational visibility. We look forward to your approval and are prepared to begin immediately upon confirmation.

Summary:

The total cost for the complete network implementation for MediFleet, which includes all hardware, software, licensing, structured cabling, and professional services, amounts to $93,850. This total comprises $60,850 allocated for equipment and materials and $33,000 for installation, configuration, and training. This investment will provide a secure, scalable, and fully compliant infrastructure specifically designed to support MediFleet's mission-critical logistics operations, ensuring reliable connectivity, data protection, and future growth.